Dynamic Chemicals





Offerings

Water Treatment Chemicals
Cooling Tower Treatment
Boiler Water Treatment
RO Membrane Treatment
Polyelectrolytes
Efficient Microbes

Fuel Additives and Fireside Chemical

Industrial Water Treatment
Softening
De alkalisation
Demineralisation
Mixed Bed resin
Nuclear Industry

Ion Exchange Resins
Groundwater Purification
Adsorbent Resin
Chelating Resin
Wastewater Treatment Resin
Resin for other Potential Applications

Construction Chemicals
Concrete Admixtures
Surface Treatment
Water Proofing
Sealants
Hardener

All Housekeeping Products
Toilet Cleaner
Air Freshners
Liquid Hand Soap
Dish Wash
Phenyl
Glass Cleaner

A Touch of perfection

Boiler & Its Types

There are mainly two types of boiler
1. Water Tube Boiler
2. Fire Tube Boiler

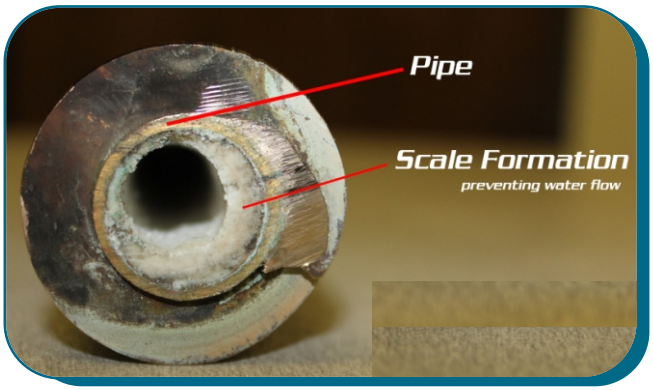


Water Tube Boiler:- A steam boiler in which water to be heated circulates in tubes exposed to fire and enveloped by hot gases.

Fire Tube Boiler:- In fire tube boiler, there are numbers of tubes through which hot gases are passed and water surrounds these tubes.

Main Components of Combustion boiler:-

Burner, Combustion chamber, heat exchanger, Supply lines and Return line, Circulatory pumps

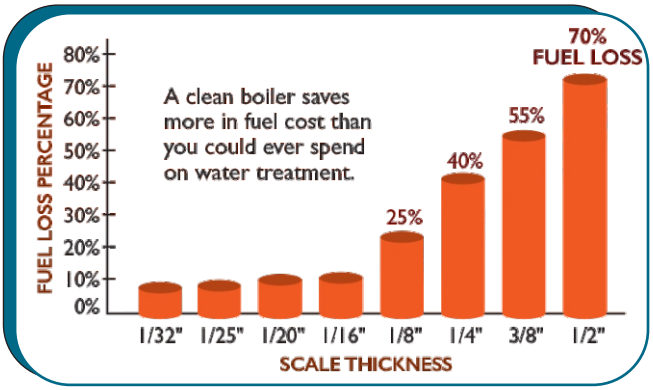


Boiler Scales

Scales are formed out of salts. Those salts come inside the boiler through water in form of soluble solids and are precipitated when the concentration level increases because of evaporation. Scales are basically crystalline and more difficult to remove.
Scales are maily made up of calcium carbonate ($CaCO_3$), Calcium sulphate ($CaSO_4$), Barium Sulphate ($BaSO_4$), Calcium Fluoride (CaF_2), Calcium Phosphate [$Ca_3(PO_4)_2$].

Disadvantages of Boiler Scales

1. **Wastage of fuel** :- Scales have a thermal conductivity so the rate of heat transfer from boiler to inside water is greatly reduced. In order to provide a steady supply of heat to water, excessive or overheating is done and this causes increase in fuel consumption.
2. **Decrease in Efficiency of Boiler** :- The excessive scale formation may cause clogging of Boiler tubes which decreases the efficiency of boiler .
3. **Danger of Explosion** :- The scale formation also leads to the uneven expansion of the boiler metal. Due to this thick scales crack and water comes suddenly in contact with overheated iron plates. This results in the formation of a large amount of steel suddenly which turn into high pressure. The sudden high pressure thus developed may lead to explosion of boiler.
4. **Corrosion**:- There are certain salts formed from scale, corrodes the tubes, thus there is steal greater corrosion of the boiler tubes.
5. **Boiler Becomes Unsafe** :- Overheating of boilers occurs due to scale formation and metal become soft and weak. This makes the us of Boiler unsafe, particularly at high pressure.



A Touch of perfection

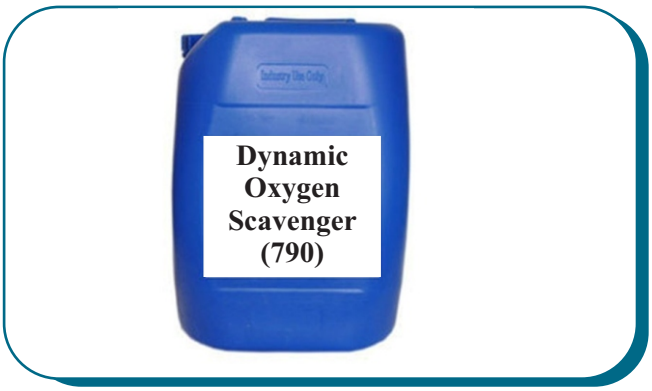
Removal of Scales & Sludges

Scales can be removed with the help of Antiscalant Chemical, Blow down Operations, EDTA Solution.

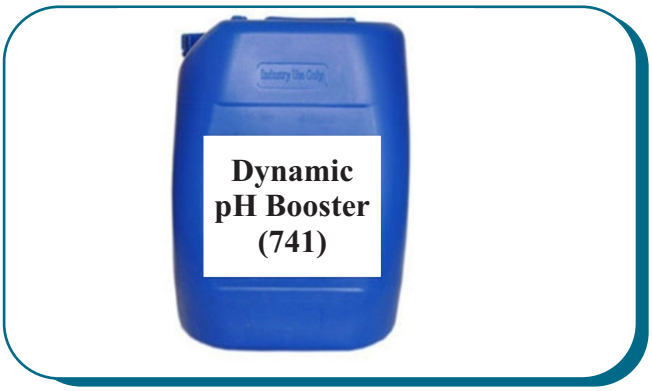


1. **Antiscalant Chemicals** :- Antiscalants are a family of chemicals like Antiscalant, oxygen scavenger, pH Booster designed to inhibit the formation and precipitation of crystalized mineral salts that form scale.

Antiscalant compound is a dianionic polyelectrolyte(DAPE), or more particularly poly[disodium 3-(N,N-diallylamino)propanephosphonate]



2. **Oxygen Scavenger** :- Oxygen scavengers are used to remove dissolved oxygen from the boiler feed water and boilers. Dissolved oxygen in the boiler is very corrosive at high temp. and pressure. This may lead to localized patting and premature boiler tube failure or excessive feed water leaks.



3. **pH Booster** :- pH Booster is a rigorous blend of alkalinity blunder, softening agent and synthetic organic sludge conditioner. It counteracts the fall in pH in the permeate water due to removal of salts. It is highly compatible for portable applications.

Water Solubility : Completely Soluble

Appearance : Clear Liquid

4. **Fireside** :- Dynamic chemical provides a series of fuel additives and fireside chemicals for liquid and solid fuel fired boilers where the impact of clinker formation can result in frequent cleaning, increased downtime and increase in specific fuel consumption.



RO(Reverse Osmosis)

Reverse Osmosis (RO) :- Reverse Osmosis is widely used in the residential and commercial water filteration system, used to remove hardness of water and make it suitable for used in feed water tanks in industries.
RO is now being used in various applications including selective separation, purification and concentration process.



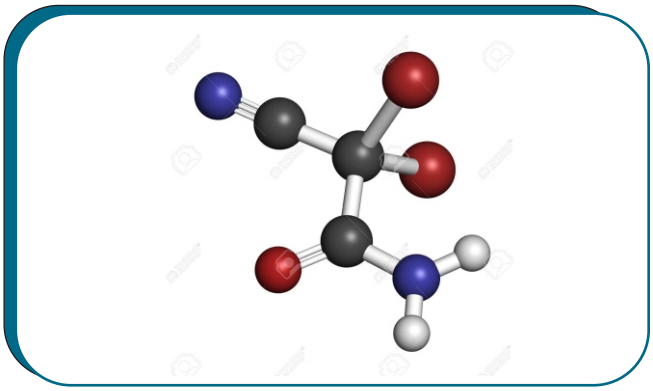
Ro Scales:- Common foulants and scalants include calcium carbonate, calcium sulphate, barium sulphate, strontium sulphate and other organic contaminants.

RO Antiscalants :- Antiscalant is a pretreatment water additive for reverse osmosis that is highly effective in preventing the membranes from scaling. The delay allows the water to pass through the membrane before any chemical reaction, in which scale can form, occurs.

pH Booster :- Dynamic's pH Booster for RO is rigorous blend of alkalinity builder, softening agent and synthetic organic sludge container. It counteracts the fall in pH in the permeate water due to removal of salts. It is highly compatible and ready to use composition for potable applications.

Cooling Tower





Cooling Tower :- A Cooling tower is a specialized heat exchange in which air and water are brught into direct contact with each other. i.e, cooling towers are essentially large boxes designed to maximize evaporation of water.

Cooling Tower Antiscalant:- Organic dispersants These cooling tower water treatment can be used to prevent fouling and th accumulation of biofilms. Dynamic chemical's Antiscalant can be used as a disinfectant to display microorganisms and prevent recurring growth.

Biocides :- Biocides are used to control bacteria in cooling tower system. Under deposit corrosion occurs when bacteria occuring within the water system. Under this bacteria corrosion occurs. The microorganisms killed by Biocides are through electrochemical process of oxidation with the help of oxidising agents.

Anticorrosion Chemicals:- Corrosion control can be done with the use of Corrosion inhibitors and by removal of corrosive oxygen from the water by chemical reaction. i.e,

1. Passivating (anodic) inhibitors.
2. Precipitating (Cathodic) inhibitors.

Descaling Process

Descaling : Descaling is the process of removing scale from equipment such as boilers and heat exchangers. It involves the removal of thick layers of oxides formed on metals at elevated temperatures. Descaling safely and effectively removes Limescale; a hard, off-white, chalky deposit that can impair the operation of various components.

SCALE DENSITY **Vs** ACTION PLAN

Scale Density Ranges and Required Action	
Scale Density g/ft2 or (mg/cm2)	**Recommended Action**
< 23 (25)	No action required
23 – 70 (25 – 75)	Chemically clean within one year
70 – 93 (75 – 100)	Chemically clean within three months
> 93 (100)	Chemically clean before further operation

PHILOSOPHY OF TREATMENT

Dynamic chemicals has been manufacturing safe chemical solutions for years, so we are familiar with all shapes, sizes and boiler configurations. The procedure outline below is brief and simple for most boiler cleanings.

1. Shut down boiler.
2. Blow down with pressure still on.
3. Drain and flush unit.
4. Close steam valve.
5. Add DYNAMIC CHEMICALS SOLUTION, and then top off with water.
6. Circulate through the drain or low point and return from the pressure relief valve or high point of the boiler to a vented container.
7. Circulate for 14-18 hours depending on volume.
8. Open steam valve.
9. Drain, flush, re-fill and start up.

DESCALING CHEMICALS

Conditioning Chemical
D-Anti Scalent Chemical
Neutralizer Chemical



Housekeeping Chemicals



(5Lt.)

Floor cleaner concentrate is a fully formulated cleaner-cum-disinfectant for cleaning and sanitising all surfaces.



(5Lt.)

1. D-Star Air Freshener is a water based air freshener featuring a pleasing floral perfume which deodorizes guestrooms, banquet halls and office rooms.
2. Long elastic Fragnance.



(5Lt.)

1. White/Black Disinfectant fluid phenolic type.
2. A Strong germicide and Antinsect.
3. Pine Oil based.
4. Super Hygiene.



(5Lt.)

Regular cleaning of windows, glass display cases and mirrors with D-Star D3, leaves the surface clean and streak-free. Concentrated for economy of use.



(5Lt.)

1. D-Star Dish Washer D9 is a concentrated manual dishwashing liquid, effective in removing all grease and food soils.
2. Contains a blend of actives to eliminate grease and dried-on food stains.



(5Lt.)

Acid based cleaner for use on most bathroom surfaces. Removes lime and scale from porcelain and metal fixtures such as sinks, baths, toilets. Cuts through soap scum and body oil.



(500ml.)

1. D-Star Toilet Cleaner is an effective toilet cleaner specially formulated to remove lime-scale deposits and stubborn stains.
2. It leaves toilet bowls and urinals sparkling clean



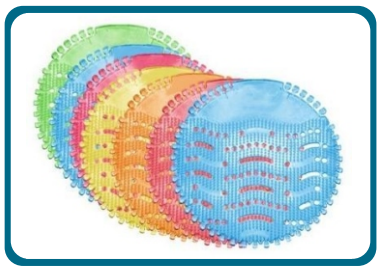
(500ml.)

- Herbal
- Antiseptic
- Biodegradable
- Best Fragnance.





Garbage Bag All Types



Urinal Screen Pad



Floor Mops Wipers



Aroma Oil
(Lemon Grass)



Soap Dispenser



Sodium Hypochlorite Solution



Napthalene Balls



Formalin

A Touch of perfection

INTEGRALLY SUSTAINABLE, HIGH PERFORMING SOLUTIONS

HARMLESS FOR HUMAN, HEALTH AND NATURE

FLOOR & SURFACES



SANITARY ROOMS



KITCHENS



ON PREMISE LAUNDRY



INDUSTRY



PERSONAL CARE



MANUFACTURER

&

SERVICE PROVIDERS

KEY SECTORS

KEY SECTORS

PUBLIC INSTITUTIONS



HEALTH CARE



INDUSTRY



LODGING



LEISURE



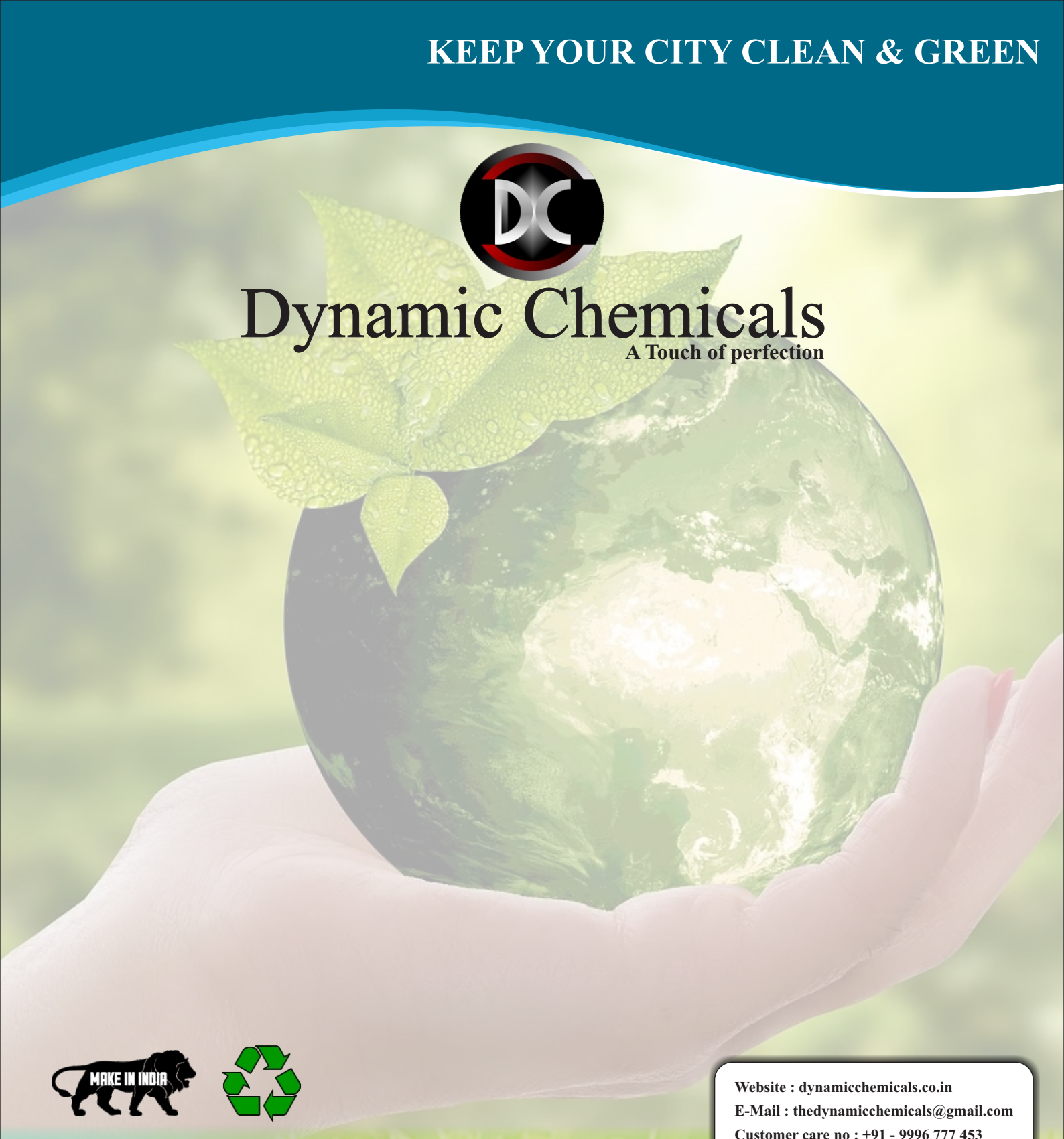
FOOD SERVICE



KEEP YOUR CITY CLEAN & GREEN

Dynamic Chemicals

A Touch of perfection



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स्वच्छ भारत

स्वच्छ भारत अभियान

एक कदम स्वच्छता की ओर